

Certificate of Analysis

PRODUCT DETAILS

Product Name	Secretome
Batch	
Manufacturing Date	
Expiry Date	
Dosage Form	Clear-colourless solution
Storage Temperature	4 – 8°C

TEST	SPECIFICATION	TEST RESULTS	REMARKS
Infectious Diseases Screening	HBsAg; HCV Ab; HIV1/2 Ag/Ab; CMV Ab; RPR	Not Detected / Non-Reactive	QUALITY CONTROL APPROVED
Sterility	No cloudiness of the test media	Negative	QUALITY CONTROL APPROVED
Bacterial Endotoxin Test (LAL)	Endotoxin sensitivity 0.125 EU/mL	Negative	QUALITY CONTROL APPROVED
Mycoplasma	Mean of absorbance <0.05	Negative	QUALITY CONTROL APPROVED
Nanoparticle Tracking Analysis (NTA)	Secretome particle size range 40nm – 1000nm	50 x 10 ⁹ particles/mL	QUALITY CONTROL APPROVED

The cell product delivered by Malaysian Genomics Resource Centre Berhad is manufactured and tested for research purpose only. Product is processed in a facility operating under current Good Manufacturing Practice (cGMP) guidelines in accordance with PIC/S standards.

Storage recommendation:

- 1) Refrigerate the vials in 4-8°C at all times upon received
- 2) Keep away the vials from direct heat or sunlight
- 3) Avoid prolong exposure in room condition
- 4) It is recommended to place the vials in freezer for prolonged storage

Authorised Signature:

**Quality Control Department
Date:**

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Supporting Information

STERILITY TEST ^{6, 8}

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.
Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

BACTERIAL ENDOTOXIN TEST (LIMULUS AMOEBOCYTE LYSATE, LAL) TEST ^{3,6}

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

MYCOPLASMA TEST ^{4,6}

Negative findings for mycoplasma species *M. hyorhinitis*, *M. arginini*, *M. fermentans*, *M. orale*, *M. Pirum*, *M. hominis*, *M. salivarium* and *A. laidlawii*.

NANOPARTICLE TRACKING ANALYSIS (NTA) ^{7,8}

Established method for the direct and real-time visualization and analysis of the nanoparticles size distribution and concentration in liquids.



REFERENCES

1. Bacterial Endotoxins, European Pharmacopoeia 5.0; <https://gmpqa.com/validation/Method/LAL/EUPHARMACPOEIA.pdf>
2. US Pharmacopeia <ES> Mycoplasma Tests
3. ICH Harmonised Tripartite Guideline, Specifications: Test Procedures and Acceptance Criteria for Biotechnological/ Biological Products Q6B.
4. US Pharmacopeia <TS> Sterility Test
5. Neha Chopra, Brahm Dutt Arya, Namrata Jain, Poonam Yadav, Selma Wajid, Surinder P. Singh, Sangetta Choudhury, 2019, Biophysical Characterization and Drug Delivery Potential of Exosomes from Human Wharton's Jelly-Derived Mesenchymal Stem Cells.
6. Leila Dareshtendi, Shih Shah, Tahereh Jafari, Maumita Bhattacharjee, Deandra Momah, Nikoo Savesh-Shemshaki, Kevin W-H Lo, and Cato T. Laurencin, Emergence of the Stem Cell Secretome in Regenerative Engineering (2021)
7. Chettimada, Sukrutha, Lorenz, David R., Misra, Vikas, Dillon, Simon T., Reeves, R. Keith, Manickam, Cordelia, 2018, Exosome markers associated with immune activation and oxidative stress in HIV patients on antiretroviral therapy.

